

Lecture 2

Spatial modelling in inlabru

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- Why spatial modelling?
- Why is spatial modelling computationally expensive?
- Different data types:
 - Modelling in discrete space – areal data
 - Modelling in continuous space – geo-referenced data
 - Modelling continuous space – spatial point process data
- representation of data structures in R

Spatial modelling

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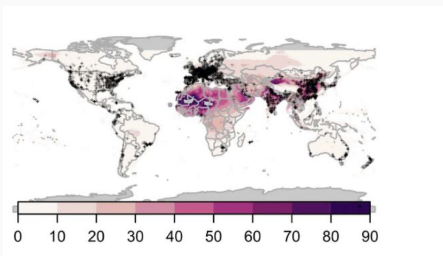
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We will see that `inlabru` can help...

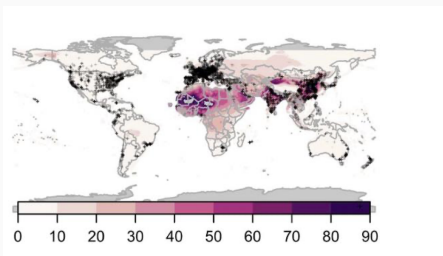
Example: Global PM 2.5



Exposure to air pollution; particulate matter $< 2.5\mu m$ (PM 2.5)

- linked to poor health outcomes; responsible for 3 million deaths worldwide each year
- observations potentially **not independent**
- sparsely measured
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spatial dependence + complex observation processes

Spatial modelling: why necessary?

- standard models assume independent observations
- spatial data are (typically) not independent
- two nearby observations are similar $\rightarrow$ not providing independent information

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There are many different ways of “being dependent”, hence these **components** and their properties vary with each application.

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- INLA (and hence `inlabru`) much faster...

Spatial modelling: aims

aims vary among different studies, e.g.:

- describing spatial patterns and structures
- understanding spatial patterns and structures
- linking data observed in space to covariates while accounting for spatial autocorrelation
- predicting into unobserved locations

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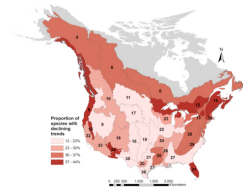
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aims also vary with different types of spatial data...

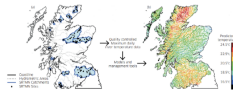
Types of spatial data

Areal data



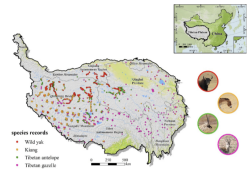
Map of bird conservation regions (BCRs) showing the proportion of bird species within each region showing a declining trend

Geostatistical data



Scotland river temperature monitoring network

Point pattern data



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Discrete space:

- Data on a spatial grid (areal data)

Continuous space:

- Geostatistical (geo-referenced) data
- Spatial point patterns (point-referenced data)

Model components are used to reflect spatial dependence structures in discrete and continuous space.

Discrete space: areal data

- Data on a (regular or irregular) spatial grid
- Examples: number of individuals in a region, average rainfall in a province
- Originally point or geostatistical data; gridded for practical reasons

Observed response(s): Measurement associated with each grid cell or areal unit

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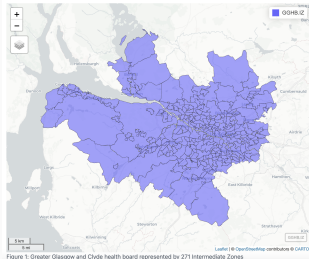
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here: spatial structure represented by Gauss Markov random field (GMRF)

the data we will use

respiratory hospitalisations in Glasgow:

- areal data on respiratory hospitalisations
- 271 Intermediate Zones (IZ) in Greater Glasgow and Clyde health board
- covariates concern air pollution concentration and socio-economic deprivation in Scotland



Standardized Mortality Ratios (SMR)

Response here:

disease risk, estimated using **Standardized Mortality Ratios (SMR)**: for spatial areal unit i SMR is the ratio between the observed (Y_i) and expected (E_i) number of cases:

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$SMR > 1$: more observed cases than expected

⇒ **high risk area**

$SMR < 1$, fewer observed cases than expected

⇒ **low risk area**

Continuous space: geostatistical data

- phenomenon continuous in space
- measured at a finite set of locations
- examples: nutrient levels in soil, salinity in the sea

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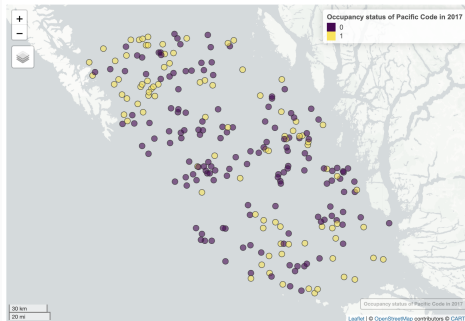
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here: spatial structure represented by Gaussian random field; approximated by a continuously indexed Gauss Markov random field

the data we will use

- Pacific Cod (*Gadus macrocephalus*)
- trawl survey in Queen Charlotte Sound



Continuous space: spatial point patterns

- Locations of objects/events in space (typically 2D)
- Examples: tree locations, animal groups, earthquakes

Observed response(s): x, y coordinates (sometimes also additional measurements; “marks”)

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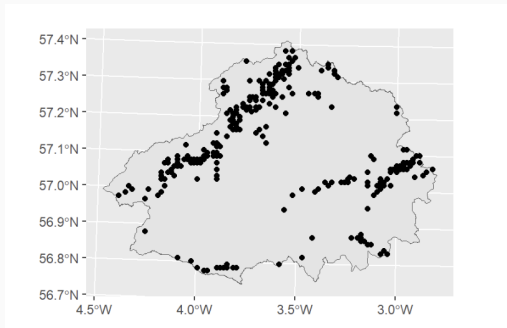
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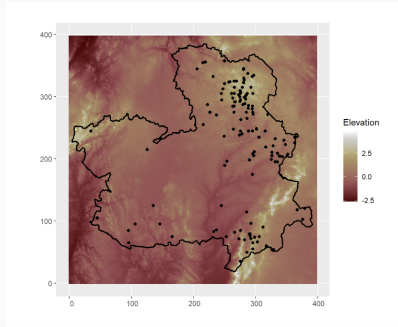
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location of sightings of the ringlet butterfly in Scotland's Cairngorms National Park



the data we will use

locations of forest fires in the Castilla-La Mancha region of Spain between 1998 and 2007



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 - GMRF can be efficiently fitted with INLA/inlabru
 - different types of models can be efficiently fitted with INLA/inlabru $\rightsquigarrow$ within the same framework
 - different data structures can be jointly modelled

Point patterns vs. geostatistical data

Point patterns:

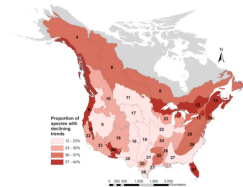
- Data format: x,y coordinates
- Optional: marks
- Aim: modelling pattern formed by locations; locations are random

Geostatistical data:

- Data format: x,y coordinates
- Measurements mandatory
- Aim: model continuous process based on measurements; locations of measurements are not random/have been chosen

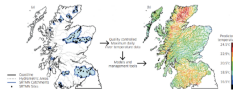
Recall: Types of spatial data

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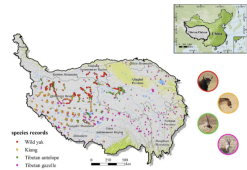
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- neighbourhood structure – neighbours in space
- e.g. we can consider random walk models in space
- these are special cases of **Gauss Markov random fields** (GMRF)
- recall: latent **Gaussian** models...

large latent Gaussian models

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- GMRFs provide a computationally efficient representation
 - we will see later that these will come in handy even for continuous data (the “SPDE approach”)

The Gaussian distribution

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- The matrix Σ describes all pairwise covariances
 $\Sigma_{i,j} = \text{Cov}[x_i, x_j]$
- Analytically tractable and well understood...

Formally

$$\pi(x) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} x^T \Sigma^{-1} x}$$

- Σ is typically large and dense
- infeasible for large n
- we need to reduce computational cost

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1. Force Σ to be sparse $\Rightarrow$ independence...
2. Force $Q = \Sigma^{-1}$ (the precision) to be sparse $\Rightarrow$ conditional independence

Example: AR(1) Process Test

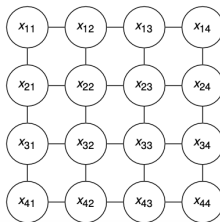
Graphical Representation

- Conditional independence structure can be represented via a graph.
- Each node corresponds to a variable x_i .
- Edges correspond to nonzero Q_{ij} .

— AR(1):



— ICAR(1):



Definition of a GMRF

A random vector x is a **Gaussian Markov Random Field (GMRF)** on a graph $G = (V, E)$ with mean μ and precision Q if:

$$\pi(x) = \frac{|Q|^{1/2}}{(2\pi)^{n/2}} \exp\left[-\frac{1}{2}(x - \mu)^T Q (x - \mu)\right],$$

and $Q_{ij} \neq 0 \Leftrightarrow \{i, j\} \in E$.

Example: AR(1) process

$$x_1 \sim \mathcal{N}\left(0, \frac{1}{1 - \phi^2}\right)$$

$$x_t = \phi x_{t-1} + \varepsilon_t, \quad t > 1, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

The covariance matrix is dense since all time points are dependent, but the precision matrix Σ^{-1} is not:

$$\Sigma^{-1} = \begin{bmatrix} 1 & -\phi & 0 & \dots & 0 \\ -\phi & 1 + \phi^2 & -\phi & \dots & 0 \\ 0 & -\phi & 1 + \phi^2 & -\phi & \vdots \\ \vdots & \vdots & -\phi & 1 + \phi^2 & -\phi \\ 0 & \dots & 0 & -\phi & 1 \end{bmatrix}$$

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- we call $Q = \Sigma^{-1}$ the precision matrix.
- key property that causes Σ^{-1} to be sparse – conditional independence
- Can we take advantage of it in other models?

(Informal) definition of a GMRF

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 - in the time series AR1 example each variable is connected only to its predecessor and successor
- ~> the precision matrix is tridiagonal, i.e. sparse
- ~> efficient computation

iCAR or Besag model:

- each region conditionally has a Gaussian distribution with
 - mean equal to the average of the neighbours and
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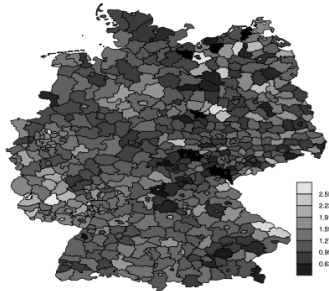
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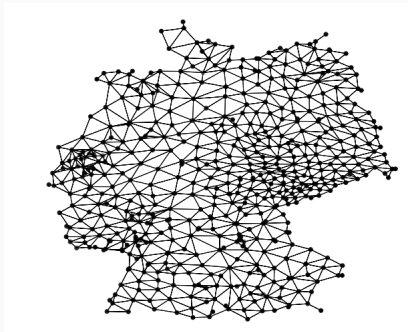


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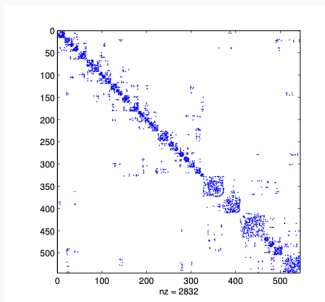


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Here – BYM (Besag-York-Mollié) model:

$$y_i | \eta_i \sim \text{Poisson}(E_i \lambda_i)$$

$$\eta_i = \log(\lambda_i) = \beta_0 + \beta_1 pm(10) + \omega_i + z_i$$

The BYM model in inlabru

The Model

$$y_i | \eta_i \sim \text{Poisson}(E_i \lambda_i)$$

$$\log(\lambda_i) = \eta_i = \beta_0 + \beta_1 \text{pm10} + u_i + z_i$$

The code

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1 # define model component
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3   space(space, model = "besag", graph = Q) +
4   iid(space, model = "iid")
5
6 # define model predictor
7 eta = observed ~ Intercept + space + iid
8
9 # build the observation model
10 lik = bru_obs(formula = formula,
11               family = "poisson",
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15 # fit the model
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neighbourhood structure

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Take-home message

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- spatial relationships are represented through a neighbourhood structure
- GMRFs are a computationally efficient representation
- these will also be relevant – and very useful – when it comes to approximating spatial structures in continuous space

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- spatial relationships are represented through a neighbourhood structure
- GMRFs are a computationally efficient representation
- these will also be relevant – and very useful – when it comes to approximating spatial structures in continuous space
- look forward to GRMFs in continuously indexed space...

Types of spatial data

Recall:

Discrete space:

- Data on a spatial grid (areal data)

Continuous space:

- **Geostatistical (geo-referenced) data**
- Spatial point pattern data

Model components are used to reflect spatial dependence structures in discrete and continuous space.

spatial data – continuous case

recall: geostatistical data

- phenomenon that is continuous in space
- examples: nutrient levels in soil, salinity in the sea, altitude
- measurements in only a finite number of locations

aim: estimate the continuous field

- continuous random variable, a **random field**: function in space with values in continuous space
- **Gaussian random field**
- characterised by mean and covariance (function)

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 - how do we best approximate the $w(s)$, GRF)?

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~> flexibly and efficiently using the **SPDE** approach

Gaussian random fields

we have a process that is occurring everywhere in space

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$\rightsquigarrow$ natural to try to model it using some sort of function (of space),

a random field $\omega(s)$ $\rightsquigarrow$ a random field is a random function...

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- What makes a function Gaussian...?

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the mesh!

the SPDE approach – more flexible models

- simple models use a simple (regular) gridding approach to approximate the continuous spatial field
 - this is easy to implement
 - however: this can be
 - **computationally inefficient** and
 - not flexible enough (complicated boundaries or domains)
- ⇒ use continuously specified finite dimensional Gaussian random fields
- ⇒ spatial field as solution to a stochastic partial differential equation (“SPDE approach”)

continuous specification

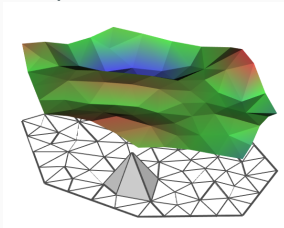
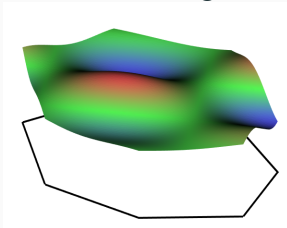
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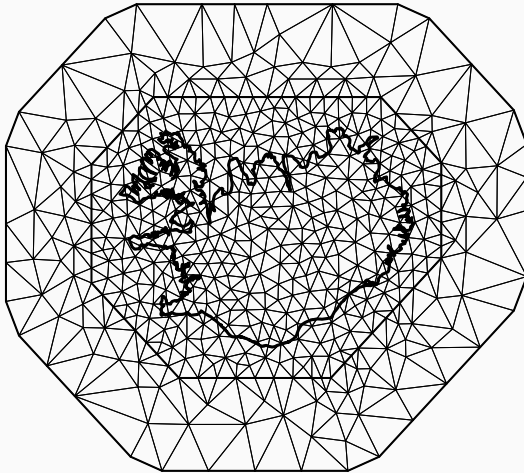
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- use a **continuous** specification of the random field model:
a finite-dimensional basis function expansion
in the “mesh” - a triangulation of space



- e.g. for point processes: no “binning” of the points
- allows computation using the **exact** positions of the points...

A mesh...



benefits of the SPDE approach...

After all those technicalities... here's the important bit:

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After all those technicalities... here's the important bit: The SPDE approach yields

- more flexible modelling; it is easier to
 - build more general models
 - work with changing observation areas over time
 - work with “funny” observation areas
- we don't need to worry about covariance functions...

Part of the magic: SPDE models are still GMRF!

⇒ we can still use INLA to fit these and it is still fast!

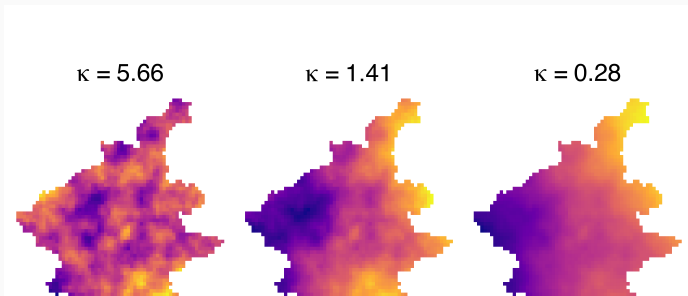
SPDE models

We call spatial Markov models defined on a mesh *SPDE models*.

SPDE models have 3 parts

- a mesh
- a range parameter ρ
- a variance parameter σ^2 (or precision parameter τ)

Different realisations of the same SPDE model with varying range parameter κ .



And now...?

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Here:

$$y(s)|\eta(s) \sim \text{Binom}(1, p(s))$$

$$\eta(s) = \text{logit}(p(s)) = \beta_0 + \omega(s) + \beta_1 \text{ depth}(s)$$

prior choice for GRFs

priors determine the smoothness of the random field

- if the field is too smooth, spurious significance
- if the field is too wiggly, overfitting

⇒ choice has to be done very carefully

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approaches (in the spatial stats literature):

- choose an arbitrary prior that your friend used and liked...
- play around – choose some arbitrary prior value and change it and check what happens until you are happy
- believe in some default prior and trust it blindly

all arbitrary...

options:

- 1) provide some standard prior distributions
 - e.g. a log-gamma distribution; user chooses parameters

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⇒ BUT: we don’t know what the ideal smoothness/wigglyness is...

as a software developer...

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we want to automate, but:

penalise something different: deviation from a base model

aim: make prior choice transparent and problem-driven

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- model component is flexible version of a **base model**
- penalise deviation from base model (using Kullback-Leibler divergence, measuring information loss)
- “shrink” towards the base model – the data really have to “convince” us that the more complicated model is necessary

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- “shrink” towards the base model – the data really have to “convince” us that the more complicated model is necessary
- the maths says that having a prior on this parameterisation will avoid overfitting
- we are penalising unnecessary model complexity
- also: interpretable priors and **flexible modelling**

make prior choice transparent

penalise deviation from a **base model** in our context...

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The base model:

- model with intercept and covariates, no (or rather a very flat) Gaussian random field

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⇒ the approach is conservative about including a random field – penalises overfitting

⇒ we set a prior on the spatial scale that we consider overfitting

⇒ we have another prior that reflects how confident we are about this scale

prior value that reflects the spatial scale that we consider overfitting

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How do we do this?

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Use `inla.spde2.pcmatern` with prior parameters:

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- i.e. we are pretty sure that a range of this size or smaller is overfitting

Take-home message

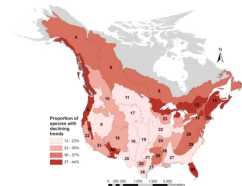
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Take-home message

- Gaussian random fields provide components that reflect spatial structure in continuous space
- the SPDE approach allows us to approximate these efficiently and flexibly
- these models are still GMRFs
 - ~> computationally efficient representation
- pc-priors are also relevant for spatial models, e.g. to avoid overfitting

Recall: Types of spatial data

Areal data



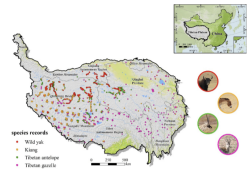
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Continuous space: spatial point patterns

- Locations of objects/events in space (typically 2D)
- Examples: tree locations, animal groups, earthquakes

Observed response(s): x, y coordinates (sometimes also marks)

Point patterns vs. geostatistical data

Point patterns:

- Data format: x,y coordinates
- Optional: marks
- Aim: model locations as random

Geostatistical data:

- Data format: x,y coordinates
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spatial point processes – what are they?

models of spatial patterns:

⇒ modelling **locations and properties (“marks”)** of objects,
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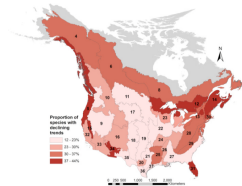
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~> some methodology that has been developed for other data structures is not always available for point processes (yet) – e.g. model comparison is difficult (see lecture 10)

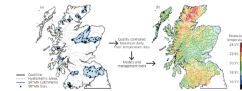
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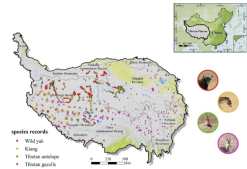
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models of spatial patterns:

⇒ modelling **locations and properties** (“marks”) of objects,
events, individuals in space and time

aim: understanding mechanisms

that generated the pattern

- examples:
 - cancer cells
 - plants or animals
 - earthquakes
 - terrorist attacks

spatial point processes – why are they complex?

Point patterns can be modelled by a **point process** N

- assigns a count of points to every subset in W ; a **measure**;
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- is practically relevant and realistically complex,
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~> making the very theoretic point processes relevant in practice

~> some methodology that has been developed for other data structures is not always available for point processes (yet) – e.g. model comparison is difficult (see lecture 10)

- analysis of **marked point patterns** is more interesting
- relevance: provides deeper insight into the processes that are causing the pattern than an analysis of unmarked point patterns
- marks may be
 - **qualitative**, i.e. the pattern is multivariate and consists of several types of points, e.g. different species, ages and size classes, or
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locations of eucalyptus trees in a koala reserve; diameters of the circles reflect the palatability of the trees' leaves

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diameters of the circles reflect the number of times a koala was observed on a tree in a given time period

a model for spatial patterns...



question:

- are the daisies randomly distributed in the lawn of our garden???

issues

- what do we mean by “random”? formal description?
- what if they are not random?
- how should we describe and model non-random patterns?

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- (1) *Poisson distribution of point counts*: number of points of in any set A follows a Poisson distribution with mean $\lambda \|A\|$
- (2) *Independent scattering*: number of points of in k disjoint sets are independent of each other

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- $\lambda \cdot \|A\| = \mathbb{E}(N(A))$ for all sets A

Poisson process – more interesting models...

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Cox process or

(3) non-independence: interaction among the points

Gibbs process [not discussed here]

constant intensity λ of the homogeneous Poisson process is replaced by an intensity function $\lambda(x)$

Point process models

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- (1) *Poisson distribution of point counts.* number of points N in any set B has a Poisson distribution with mean $\int_B \lambda(x) dx$
- (2) *Independent scattering.* The random numbers of points of N in k disjoint sets are independent random variables, for arbitrary k .

Cox process:

- intensity function is replaced by a **random field** $\Lambda(x)$ with non-negative values

Point process models – Cox processes

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example:

log-Gaussian Cox process (LCP):

$$\log(\Lambda) = Z(s),$$

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↪ flexible, but hard to fit!

Point processes in inlabru

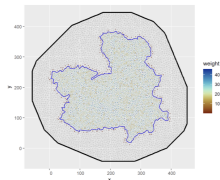
The LGCP Model

$$p(y|\lambda) \propto \exp\left(-\int_{\Omega} \lambda(s)ds\right) \prod_{i=1}^n \lambda(s_i)$$
$$\eta(s) = \log(\lambda(s)) = \boxed{\beta_0} + \boxed{x(s)} + \boxed{\omega(s)}$$

The code

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1 # define model component
2 cmp = ~ Intercept(1) + elev(elev_raster, model = "linear") +
3   space(geometry, model = spde_model)
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6 # define model predictor
7 eta = geometry ~ Intercept + elev + space
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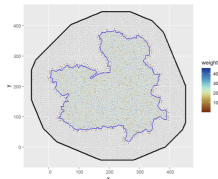
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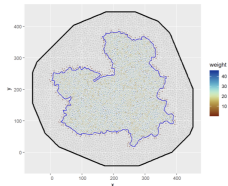
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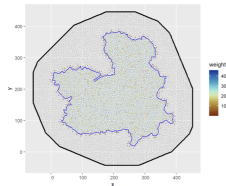
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Intuitively:

Can account for additional “spatial over dispersion” that is structured- i.e. patterns that covariates cannot explain

and what about marked point processes?

recall: marked point processes

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- see lecture tomorrow on joint modelling

What happens next

- Practical 5 continued: spatial point process fitting
- homogeneous Poisson process
- inhomogeneous Poisson process
- Log-Gaussian Cox process

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